

Sri Chaitanya IIT Academy.,India.

A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -2(Adv-2019-P2-Model)** **Date: 15-04-2026**

Time: 02.30Pm to 05.30Pm **GTA-24** **Max. Marks: 186**

15-04-2026_Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT_Jee-Adv(2019-P2)_GTA-24_Syllabus,

PHYSICS : TOTAL SYLLABUS

CHEMISTRY : TOTAL SYLLABUS

MATHEMATICS : TOTAL SYLLABUS

Rankers Academy JEE
Sri Chaitanya Educational Institutions INDIA

Name of the Student: _____ **H.T. NO:** | | | | | | | | |

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JEE-ADVANCE-2019-P2-Model

Time: 3:00Hr's **IMPORTANT INSTRUCTIONS** **Max Marks: 186**

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 8)	Questions with Multiple Correct Choice **+1 partial marks**	4	-1	8	32
Sec – II(Q.N : 9 – 14)	Questions with Numerical Value Answer Type	3	0	6	18
Sec – III(Q.N : 15 – 18)	Matching Type	3	-1	4	12
Total				**18**	**62**

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 19 – 26)	Questions with Multiple Correct Choice **+1 partial marks**	4	-1	8	32
Sec – II(Q.N : 27 – 32)	Questions with Numerical Value Answer Type	3	0	6	18
Sec – III(Q.N : 33 – 36)	Matching Type	3	-1	4	12
Total				**18**	**62**

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 37 – 44)	Questions with Multiple Correct Choice **+1 partial marks**	4	-1	8	32
Sec – II(Q.N : 45 – 50)	Questions with Numerical Value Answer Type	3	0	6	18
Sec – III(Q.N : 51 – 54)	Matching Type	3	-1	4	12
Total				**18**	**62**

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PHYSICS **Max Marks: 62**

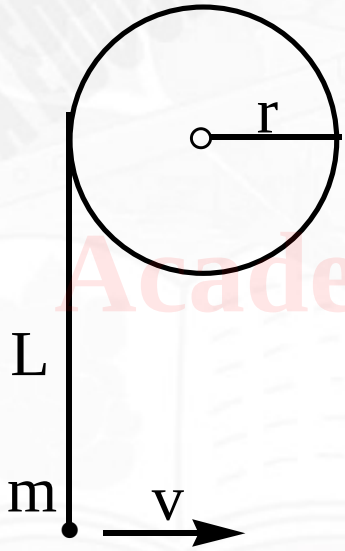
SECTION – I
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct. **Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -1 in all wrong cases**

Section 1 (Max Marks: 32)

- Section 1 contains eight questions.
- Each Question has Four Options and Only One of these four will be the correct answer.
- For each question, choose the option corresponding to the correct answer
- The Marking scheme to evaluate the Answer to each question:
  Full Marks: +4 (If the answer is correct)
  Partial Marks: +3 (If all the four options are correct but only 3 are chosen)
  Partial Marks: +2 (If THREE or More options are correct but only 2 correct answers are chosen)
  Partial Marks: +1 (If TWO or More options are correct but only 1 correct answer is chosen)
  Zero Marks: 0 (If the question is unanswered)
  Negative Marks: -1 (In all other cases)

1. A horizonal disk of radius $r = 0.2m$ is fixed onto a horizontal frictionless table. One end of a massless string of length $L = 0.8m$ is fixed to the perimeter of the disk, while the other end is attached to an object of mass $m = 0.6kg$, which stands on the table as shown. The object is then given a velocity of magnitude $v = 0.4m/s$ in a direction perpendicular to the string. Choose correct statement(s)



A) The time when the object will hit the disc is 3 seconds.

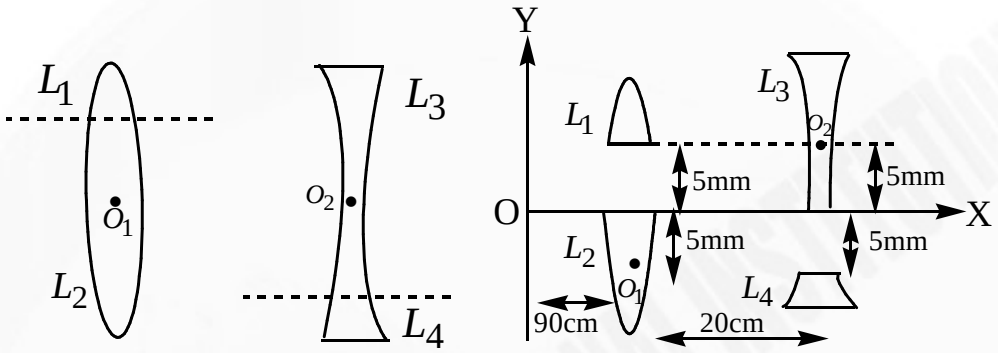
B) The time when the object will hit the disc is 4 seconds

C) Tension in the string as a function of time, t is $T = \sqrt{\dfrac{mv^2}{L^2 - 2rvt}}$.

D) Tension in the string as a function of time, t is $T = \sqrt{\dfrac{mv^2}{L^2 - rvt}}$.

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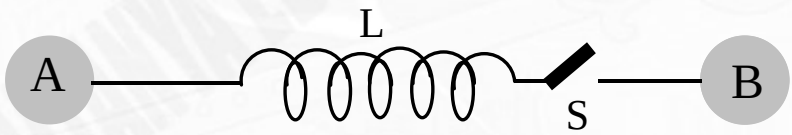
2. A convex lens is divided into two parts at a distance 5 mm from the optic centre and two parts are placed at a separation 5 mm as shown. A concave lens is also divided into two parts but in opposite sence that of convex lens. The focal lengths of convex and concave lenses are 30 cm and 50 cm (numerically). The possible coordinates of real images are (O is point object kept at origin as shown, O_1 & O_2 are optic centers of lenses)



A) $(160cm, -1.5cm)$ **B)** $(135cm, -0.75cm)$

C) $(160cm, -0.5cm)$ **D)** $(160cm, -2cm)$

3. Two metal spheres A and B of radii R are placed at a very large distance from each other, and they are connected by a coil of inductance L, as it is shown in the figure. Sphere A is loaded with electric charge. Switch is closed at t = 0



Choose correct option(s)

A) The time after which the charge on sphere A again become original charge is $2\pi\sqrt{2\pi\varepsilon_0 LR}$.

B) The time after which the charge on sphere A again become original charge is $\pi\sqrt{2\pi\varepsilon_0 LR}$.

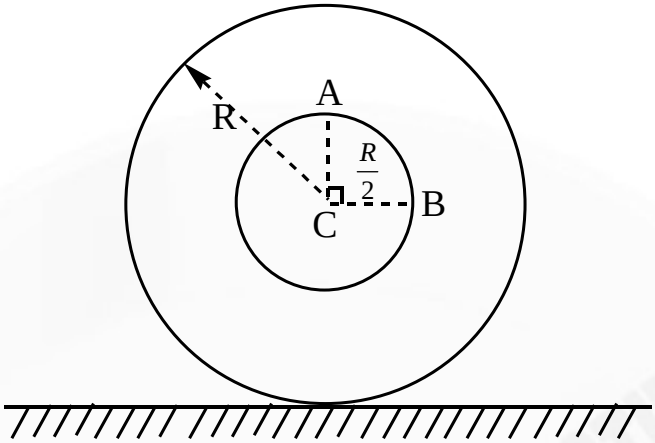
C) The time after which the charge on sphere A decreases to half is $\pi\sqrt{2\pi\varepsilon_0 LR}$.

D) The time after which the charge on sphere A decreases to half is $\frac{\pi}{2}\sqrt{2\pi\varepsilon_0 LR}$.

4. A uniform laminar disc rolls without slipping on a rough horizontal surface with uniform angular velocity. Point c is the centre of disc, A and B are two points on the disc at a distance of $\frac{R}{2}$. Which of the following statements are incorrect:

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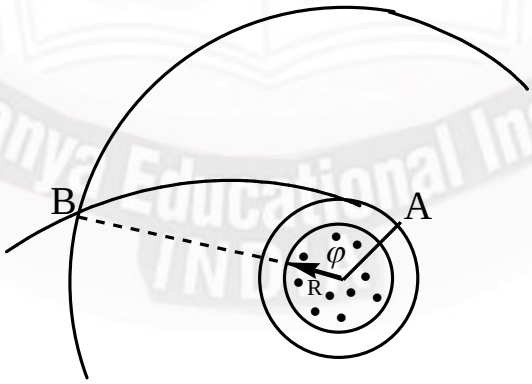

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A) The acceleration of A w.r.t B is uniform
B) The acceleration of A is uniform.
C) There are many points on the disc whose acceleration is equal to of acceleration of A.
D) No two points on the disc have same acceleration

5. Mark the correct statement(s) regarding nuclear reactions.
A) For an exothermic reaction if Q value is +12.56 MeV and the KE of incident particle is 2.44 MeV, then the total KE of products of reaction is 15.00 MeV.
B) For an exothermic reaction, if Q value is +12.56 MeV and KE of incident particle is 2.44 MeV, then the total KE of products of reaction is 12.56 MeV.
C) For an endothermic reaction, if we give the energy equal to |Q| value of reaction, then the reaction will be carried out.
D) For an exothermic reaction, the average BE per nucleon of products should be greater than the average BE per nucleon of reactants

6. In the interior of a long straight coil of radius $R = 2cm$, the magnitude of the magnetic induction decreases uniformly from $B = 0.8Vs/m^2$ to zero in a time interval of $\Delta t = 10^{-1}s$.



Choose the correct option(s):

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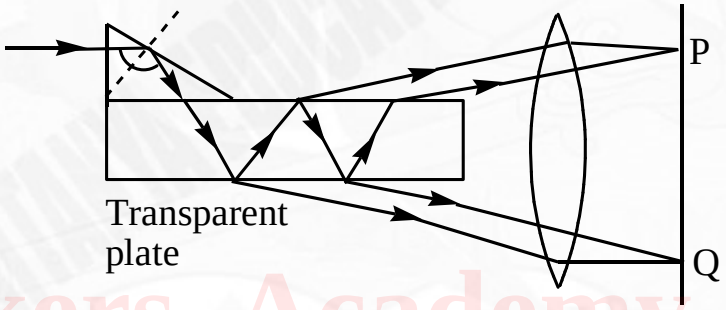
A) Initial acceleration of an electron at rest at a distance $r_A = 3cm$ from the axis of coil in approximately $8.28\times10^9\, m/\sec^2$.

B) Initial acceleration of an electron at rest a distance $r_A = 3cm$ from the axis of coil in approximately $9.38\times10^9\, m/\sec^2$.

C) Speed gained by the electron as it gets to point B while covering an angular displacement of $\phi = 120^0$ is approximately 34.3 km/s

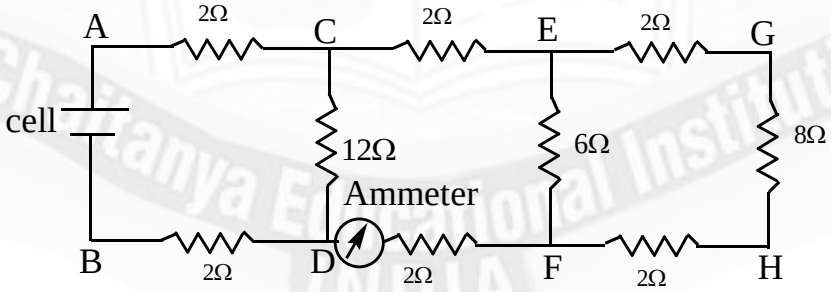
D) Speed gained by the electron as it gets to point B while covering an angular displacement of $\phi = 120^0$ is approximately 32.4 km/s

7. In an experiment with Lummer Gehrecke plate, the two coherent beams of light, caused by multiple reflections inside the transparent plate of refractive index $\mu = 1.54$, reach the points P and Q on the screen. The net path difference between the two beams reaching either at P or Q is $\Delta x = 5000\, nm$. Which of the wavelength in the visible range $(\lambda = 390\, nm \text{ to } \lambda = 780\, nm)$ is/are most likely to produce a constructive interference (a maximum) at the point P as well as at Q on the screen?



A) 416.67 nm **B)** 555.56 nm **C)** 625.0 nm **D)** 666.70 nm

8. In the circuit shown the current in the 8Ω resistance across G and H is I = 0.5 ampere. The ammeter is ideal. The internal resistance of the cell is 0.8Ω. Choose correct option(s).



A) Reading of the ammeter is 1.5 ampere

B) Potential difference across A and H is 13V

C) Potential difference across C and F is 9V

D) The emf of the cell is 24V

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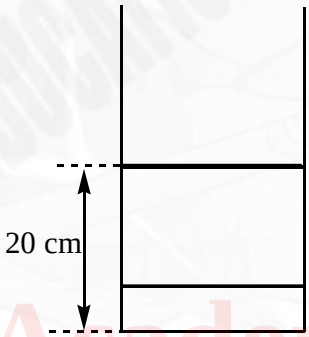
SECTION – II
(NUMERICAL VALUE TYPE)

This section contains 6 questions. Each question is numerical value type. For each question, enter the correct numerical value
(e.g. 003.2343, 6.250023, 7.000000, -0.33, -.30, 30.27000, -127.0030).
Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases. Section 2 (Max Marks: 18)

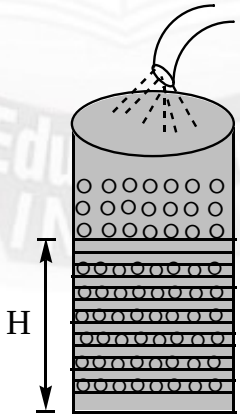
- Section two contains six (6) questions. The answer to each question is a NUMERICAL VALUE
- For each question enter the correct numerical value of the answer using the mouse and the on-screen virtual numerical keypad in the place designated to enter the answer. If the numerical values has more than two decimal places, truncate/round off the value to TWO decimal places.

Answer to each question will be evaluated according to the following marking scheme
Full Marks: +3 (If only the correct numerical value is entered)
Zero Marks: 0 (In all other cases)

9. In a cylindrical container of height 40cm two pistons enclose certain amount of gas, as is shown in the figure. The upper piston is at a height of 20 cm from the bottom of the cylinder. If the upper piston is slowly lifted by 10 cm, then the lower piston rises by 4cm. Determine the position of the lower piston in cm from bottom of cylinder, if the upper one is removed from the cylinder. The external air pressure is $10^5 Pa$, the cross section of the cylinder is $10 cm^2$ and each piston has a mass of 1kg. (Assume that the temperature is constant during the process.)

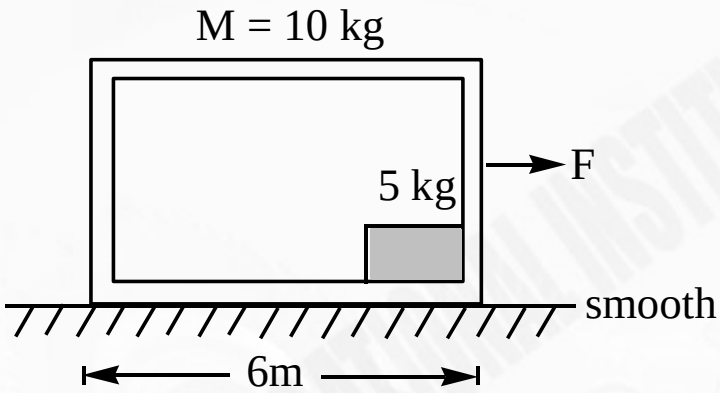


10. A cylindrical container having radius r has perforated wall. There are large number of uniformly spread small holes on the vertical wall occupying a fraction $\eta = 0.02$ of the entire area of the wall. To maintain the water level at height H in the container, water is being fed to it a constant rate Q (litre/sec). Find Q.
(Take: $H = 1.8m, r = 10cm$, $g = 10 m/s^2$)

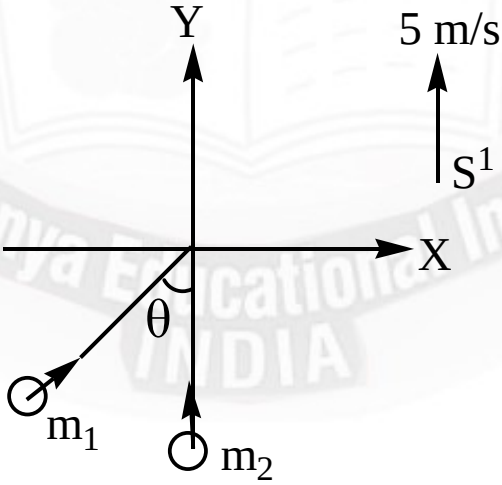


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11. Figure shows a hollow box of mass 10kg placed on a smooth surface and on its inner surface a small block of mass 5kg is placed where friction coefficient is 0.2. F(as shown in the figure) is the minimum force at which 5 kg block starts sliding on the inner surface of box. We now apply a force 2F on the box instead. Find the time in seconds after which the 5kg block will hit the other wall of box for the first time. Neglect the length of 5kg block. $\left(g = 10 m/s^2\right)$

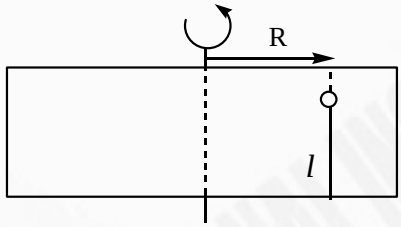


12. Two smooth spheres of the same radius, but which have different masses m_1 and m_2 collide in elastically. Their velocities before collision are $13m/s$ and $5m/s$ respectively along the directions shown in the figure in which $\cot\theta = \frac{5}{12}$. An observer S' moving parallel to the positive Y-axis with a constant speed of 5 m/s observes this collision. He finds the final velocity of m_1 to be 5 m/s along the Y-direction and the total loss in the kinetic energy of the system to be $\frac{1}{72}$ of its initial value. The magnitude of velocity of m_2 after collision with respect to observer S' is (in m/s)



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13. A closed cylindrical container with a vertical axis is completely filled with water. A plastic bead of density $\rho = 500\ kg/m^3$ and radius $r = 1cm$ is placed at distance $R = 20cm$ from the axis and is anchored to the bottom of the container by a thin thread of length $l = 16cm$. If as a result of the containers revolutions, the beads sink by $h = 4cm$, how many revolutions per second around its axis of symmetry does the container have to make? (In the final state the total content of the container rotates at the same angular speed. Calculate with $g = 10m/s^2$.)



14. A massive ball is falling down from an initial height of $h = 20m$ .With a gun held horizontally, d=50m far from the trajectory of the falling ball, at the height of $h' = 10m$, we are going to shoot at the falling ball. The bullet leaves the gun at a speed of $v = 100m/s$ .At what time is seconds after the start of the fall should the gun be fired in order to hit the falling ball with the bullet ?(The air resistance is negligible and $g = 10m/s^2$)

SECTION-III
(MATCHING LIST TYPE)

This section contains 4 questions, each having two matching lists (List-1 & List-II). The options for the correct match are provided as (A), (B),(C) and (D) out of which ONLY ONE is correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

Section 3 (Max Marks: 12)

- This Section contains two (02) List-Match sets.
- Each List-Match set has TWO (02) Multiple-choice Questions
- Each List-Match set has two lists: List 1 and List 2
- List 1 has four entries (I), (II), (III), (IV) and List 2 has six entries (P), (Q), (R), (S), (T) and (U)
- Four options are given in each multiple choice question based on List 1 and List 2 and Only ONE of these four options satisfies the condition asked in the Multiple Choice Question.

Answer to each question will be evaluated according to the following marking scheme

Full Marks: +3 (If only the option corresponding to the correct combination is chosen))

Zero Marks: 0 (If none of the options is chosen (i.e. the question is unanswered)

Negative Marks: -1 (In all other cases)

Answer Q.15 and Q.16 by appropriately matching the list based on the information given in the paragraph

A musical instrument is made using four different metal strings,1,2,3 and 4 with mass per unit length $\mu, 2\mu, 3\mu$ and 4μ respectively. The instrument is played by vibrating the strings by varying the free length in between the range L_0 and $2L_0$.It is found that in string-1 (μ) at free length L_0 and tension T_0 the fundamental mode frequency is f_0.

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List-I gives the above four string while List-II lists the magnitude of some quantity.

List-I	List-II
(I) String-1 (μ)	**(P)** 1
(II) String-2 (2μ)	**(Q)** $\frac{1}{2}$
(III) String-3 (3μ)	**(R)** $\frac{1}{\sqrt{2}}$
(IV) String-4 (4μ)	**(S)** $\frac{1}{\sqrt{3}}$
	(T) $\frac{3}{16}$
	(U) $\frac{1}{16}$

15. If the tension in each string is T_0, the correct match for the highest fundamental frequency in f_0 units will be

A) $I \to P, II \to Q, III \to T, IV \to S$ **B)** $I \to P, II \to R, III \to S, IV \to Q$

C) $I \to Q, II \to S, III \to R, IV \to P$ **D)** $I \to Q, II \to P, III \to R, IV \to T$

16. The length of the string 1,2,3 and 4 are kept fixed at L_0, $\frac{3L_0}{2}, \frac{5L_0}{4}$, *and* $\frac{7L_0}{4}$, respectively. Strings 1,2,3, and 4 are vibrated at their $1^{st}, 3^{rd}, 5^{th}$ *and* 14th harmonies, respectively such that all the strings have same frequency ,The correct match for the tension in the four strings in the units of T_0 will be,

A) $I \to P, II \to R, III \to T, IV \to U$ **B)** $I \to P, II \to Q, III \to T, IV \to U$

C) $I \to Q, II \to Q, III \to R, IV \to T$ **D)** $I \to T, II \to Q, III \to R, IV \to U$

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Answer Q.17 and Q.18 by appropriately matching the list based on the information given in the paragraph

A polarized electromagnetic plane beam $\vec{E} = 2\sqrt{\pi}\left(a\hat{i} + b\hat{j}\right)N/C \sin\left(\omega\left(t + \frac{Z}{C}\right)\right)$ is travelling along $-ve$ z direction. A polaroid is placed in its path at origin so that polaroid is in xy plane. List-I indicates pass axis & List-II emergent intensity.

	List-I (Pass axis along)		List-II (Intensity)
I.	$x = y$	P)	$\frac{49}{120} w/m^2$
II.	$x = 0$	Q)	$\frac{48}{125} w/m^2$
III.	$y = 0$	R)	$\frac{3}{20} w/m^2$
IV.	$3x = 4y$	S)	$\frac{4}{15} w/m^2$
		T)	$\frac{5}{12} w/m^2$
		U)	Zero

17. If $a = 3$ and $b = 4$, then correct match of intensity with passing axis is

A) $I \to P, II \to T, III \to R, IV \to U$ **B)** $I \to Q, II \to S, III \to R, IV \to T$

C) $I \to P, II \to S, III \to R, IV \to Q$ **D)** $I \to Q, II \to S, III \to T, IV \to U$

18. If $a = 4$ and $b = 3$, then correct match of intensity with passing axis is

A) $I \to R, II \to P, III \to S, IV \to T$ **B)** $I \to R, II \to P, III \to S, IV \to U$

C) $I \to P, II \to R, III \to T, IV \to U$ **D)** $I \to P, II \to R, III \to S, IV \to T$

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CHEMISTRY **Max Marks: 62**

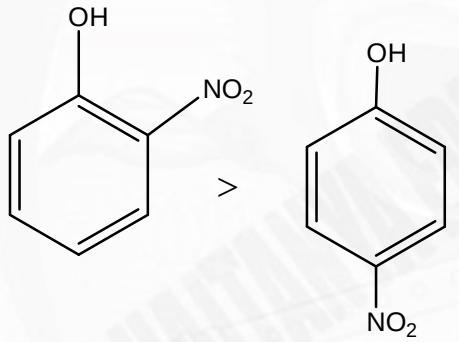
SECTION – I
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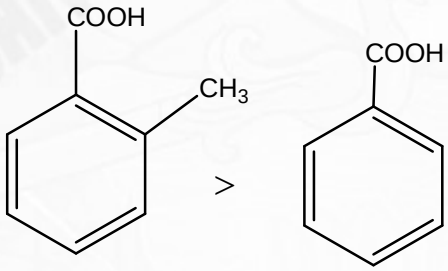
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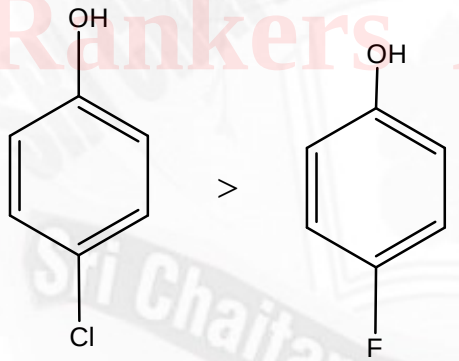
Section 1 (Max Marks: 32)

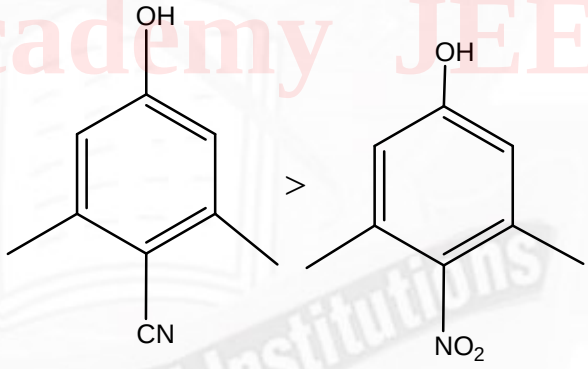
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- Each Question has Four Options and Only One of these four will be the correct answer.
- For each question, choose the option corresponding to the correct answer
- The Marking scheme to evaluate the Answer to each question:
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  Partial Marks: +1 (If TWO or More options are correct but only 1 correct answer is chosen)
  Zero Marks: 0 (If the question is unanswered)
  Negative Marks: -1 (In all other cases)

19. Which of the following is/are showing CORRECT order of acidic strength

A) 

B) 

C) 

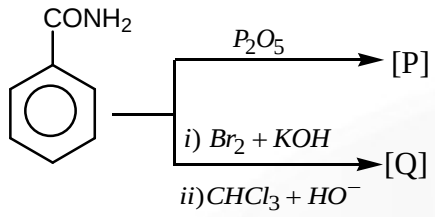
D) 

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20.



Which statement(s) is/are CORRECT?

A) Reduced product of P and Q will be metamers to each other. (Consider $LiAlH_4$ as a reducing agent)

B) By dry distillation of hydrolysed product of P with $Ca(OH)_2$ gives benzophenone.

C) One of the hydrolysis product of Q reacts with $NaNO_2 + HCl$ followed by reaction with phenol in mild basic medium gives orange red dye.

D) Electrophile involved in the formation of Q is dichlorocarbene.

21. Which of the following statements:

A) micellization is a phase transformation process.

B) micellization of soap is an endothermic process near Kraft Temperature.

C) In the micellization of soap entropy decreases.

D) As the number of carbon increases in the alkyl group of surfactant, its C.M.C decreases.

22. At 300K, the vapour pressure of an ideal solution containing 1-mole of A and 3-mole of B is 500mm Hg. At the same temp, 2 moles of B are added to this solution. The vapour pressure of solution increases by 10% of the original vapour pressure which of the following statements are correct:-

A) Vapour pressure of A in the pure state is 50mm Hg

B) Vapour pressure of B in the pure state is 650mm Hg

C) If a non volatile solute is added to final solution, its vapour pressure decreases.

D) If 4 mole of A are added in the final solution the vapour pressure becomes 350mm Hg.

23. Which of the following statements are correct?

A) An increase in pressure (caused by decrease in volume) at equilibrium results in increase in molar concentration of each gaseous substance involved.

B) An increase in pressure (caused by decrease in volume) at equilibrium results in increase in no. of moles of each gaseous substance involved.

C) For the reaction $A(s) \rightleftharpoons B(g) + C(g)$ at constant temperature, total pressure at equilibrium [of B(g) & C(g)] is not affected by changing the volume of container.

D) $H_2O(\ell) \rightleftharpoons H_2O(g)$ on temperature increase, K_P of this reaction is increased

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24. Which of the following statement(s) is/are correct? (P – pairing energy)

A) $\left[Fe(C_2O_4)_3\right]^{3-}(\Delta_0 > P)$ is a diamagnetic complex

B) Mn^{2+} compounds are more stable than Fe^{2+} towards oxidation to their +3 state.

C) $H_t - B - H_t$ angle in B_2H_6 is more than the $H_b - B - H_b$ angle

(t- terminal, b-bridge)

D) Covalence of nitrogen in N_2O_5 is 5

25. Which of the following statement(s) is/are correct?

A) The metaborate ring of $K_3\left[B_3O_6\right]$ is made up of triangular BO_3 units

B) Reaction of Pb_3O_4 with HCl involves both acid-base and redox reactions.

C) The volume of 2M HCl, in litres, required to neutralize completely 5L of 1M borax solution is 5

D) Value of 'x' in $Ca_2\ Mg_x(Si_4O_{11})_2\ (OH)_2$ is 5

26. Which of the following statement(s) is/are incorrect?

A) Azide ion is isoelectronic with CO_2, cyanamide ion, cyanate ion

B) Azide ion has two different 'nitrogen-nitrogen' bond lengths

C) Both NOF_3 and POF_3 contains similar type of bond characters between N and O in NOF_3 and between P and O in POF_3;

i.e $N = O$ *and* $P = O$ *in* NOF_3 *and* POF_3 respectively

D) Dipole moment of NF_3 is more than that of NCl_3

SECTION – II
(NUMERICAL VALUE TYPE)

This section contains 6 questions. Each question is numerical value type. For each question, enter the correct numerical value **(e.g. 003.2343, 6.250023, 7.000000, -0.33, -.30, 30.27000, -127.0030).**

Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases. Section 2 (Max Marks: 18)

- Section two contains six (6) questions. The answer to each question is a NUMERICAL VALUE
- For each question enter the correct numerical value of the answer using the mouse and the on-screen virtual numerical keypad in the place designated to enter the answer. If the numerical values has more than two decimal places, truncate/round off the value to TWO decimal places.

Answer to each question will be evaluated according to the following marking scheme

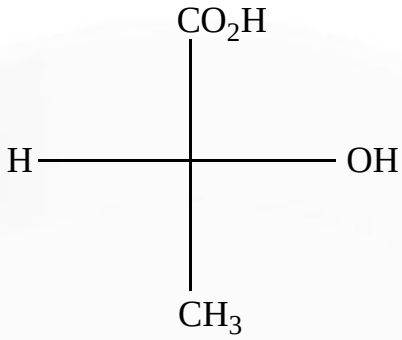
Full Marks: +3 (If only the correct numerical value is entered)

Zero Marks: 0 (In all other cases)

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27. For lactic acid, how many Fischer projection representations are possible? (structure of lactic acid is below)



28. How many are correctly matched for purification of organic compounds?

a) Distillation is used for separation of chloroform & aniline.
b) Fractional distillation for crude oil in petroleum industry.
c) Distillation under reduced pressure is used for liquid having very high boiling point.
d) Distillation under reduced pressure is used for spent-eye in soap industry.
e) Steam distillation for aniline water mixture.
f) Paper chromatography is type of adsorption chromatography.
g) Column chromatography is type of paper chromatography.
h) Thin layer chromatography is type of adsorption chromatography.

29. Enthalpy for the reaction $Ag^{+}(aq)+Br^{-}(aq)\rightarrow AgBr(s)$ is –90kJ. Magnitude of enthalpy of formation of $Ag^{+}(aq)$ and $Br^{-}(aq)$ are in the ratio 5: 6. Formation of $Ag^{+}(aq)$ is an endothermic process whereas formation of Br^{-} is an exothermic process. Enthalpy of formation of $AgBr$ is –110 kJ/mol. Calculate the enthalpy of formation of $Ag^{+}(aq)$ in kJ/mol.

30. Fixed amount an ideal gas contained in a rigid vessel $[V=24.6\ litre]$ at 1 bar is heated reversibly from $27^{0}C$ to $127^{0}C$. Determine the change in Gibb's free energy (Magnitude only) in joule. If entropy of the gas is given by $S=10+10^{-2}T\ (J/K))$

31. Dissociation constant for $Ag(NH_3)_2^{+}$ into Ag^{+} and NH_3 is 6×10^{-14}. Calculate E^{0} for the half reaction.

$Ag(NH_3)_2^{+}+e^{-}\rightarrow Ag+2NH_3$

Given, $Ag^{+}+e^{-}\rightarrow Ag$ has $E^{0}=0.8V$

$\left[\frac{2.303RT}{F}=0.06,\log 2=0.3,\log 3=0.48\right]$

Give your answer multiplying by 1000.

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32. Consider the following experiment and their observations

Experiment-I: Number of salt solutions among $CrCl_3, MnCl_2, ZnSO_4, CuSO_4$ and $Bi(NO_3)_3$ gives ppt with aq NaOH and then dissolves in excess of NaOH are 'x'.
Experiment-II: Total number of yellow coloured compounds among $K_3[Co(NO_2)_6]$, $PbBr_2, BaCrO_4, SrCrO_4, HgO.Hg(NH_2)I$, $PbCrO_4, PbSO_4$ and AgI are 'z'

Then $(x+z)$ is_____

SECTION-III
(MATCHING LIST TYPE)

This section contains 4 questions, each having two matching lists (List-1 & List-II). The options for the correct match are provided as (A), (B),(C) and (D) out of which ONLY ONE is correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

Section 3 (Max Marks: 12)

- This Section contains two (02) List-Match sets.
- Each List-Match set has TWO (02) Multiple-choice Questions
- Each List-Match set has two lists: List 1 and List 2
- List 1 has four entries (I), (II), (III), (IV) and List 2 has six entries (P), (Q), I, (S), (T) and (U)
- Four options are given in each multiple choice question based on List 1 and List 2 and Only ONE of these four options satisfies the condition asked in the Multiple Choice Question.

Answer to each question will be evaluated according to the following marking scheme

Full Marks: +3 (If only the option corresponding to the correct combination is chosen))

Zero Marks: 0 (If none of the options is chosen (i.e. the question is unanswered)

Negative Marks: -1 (In all other cases)

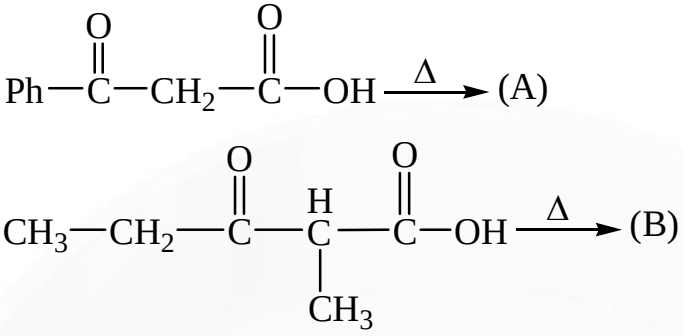
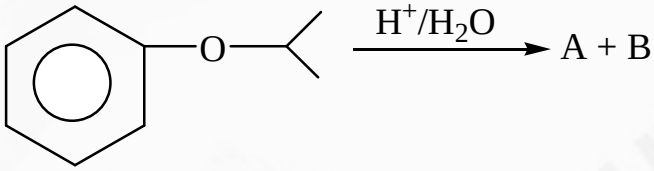
Answer Q.33 and Q.34 by appropriately matching the list based on the information given in the paragraph

Match the list

	List – I (Reactions)		**List – II (A and B is differentiated by)**
P)	$Ph-CH=O \xrightarrow[2)\ H^+]{1)\ Conc.\ \ominus OH} A + B$	1)	Neutral $FeCl_3$
Q)	$CH_3-C\equiv CH \xrightarrow[dil.\ H_2SO_4]{Hg^{+2}} A$ $CH_3-C\equiv CH \xrightarrow[2)\ H_2O_2,\ HO^-]{1)\ BH_3,\ THF} B$	2)	Fehling test

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R)		3)	$aq.NaHCO_3$
S)		4)	Iodoform test

33. Which of the following options has the correct combination considering List-I and List-II?

A) $P-4$ B) $Q-3$ C) $P-1$ D) $Q-2$

34. Which of the following options has the correct combination considering List-I and List-II?

A) $R-3$ B) $S-1,2$ C) $S-1,4$ D) $R-1$

Answer Q.35 and Q.36 by appropriately matching the list based on the information given in the paragraph

List-I		List-II (Refining method)	
I)	Zn	P)	Distillation
II)	Sn	Q)	Liquation
III)	Si	R)	Electrolytic refining
IV)	Ti	S)	Zone refining
		T)	Van Arkel method
		U)	Mond's process

35. Which of the following options has the correct combination considering List-I and List-II?

A) $I-U$ B) $I-T$ C) $III-S$ D) $III-Q$

36. Which of the following options has the correct combination considering List-I and List-II?

A) $II-S$ B) $II-T$ C) $IV-T$ D) $IV-Q$

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MATHEMATICS **Max Marks: 62**

SECTION – I
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct. **Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -1 in all wrong cases**

Section 1 (Max Marks: 32)

- Section 1 contains eight questions.
- Each Question has Four Options and Only One of these four will be the correct answer.
- For each question, choose the option corresponding to the correct answer
- The Marking scheme to evaluate the Answer to each question:
  Full Marks: +4 (If the answer is correct)
  Partial Marks: +3 (If all the four options are correct but only 3 are chosen)
  Partial Marks: +2 (If THREE or More options are correct but only 2 correct answers are chosen)
  Partial Marks: +1 (If TWO or More options are correct but only 1 correct answer is chosen)
  Zero Marks: 0 (If the question is unanswered)
  Negative Marks: -1 (In all other cases)

37. 4 unbiassed dice are thrown at random. If (a_1,a_2,a_3,a_4) is a sample point. Then which of the following option(s) is/are correct?

A) The chance that a_2 can not be greater than 2 is $\frac{1}{3}$

B) The chance that a_2 is the A.M. of a_1 and a_3 is $\frac{1}{12}$

C) The chance that $a_1 > a_2$ and $a_3 < a_4$ is $\frac{25}{144}$

D) The chance that $a_1 + a_2 = a_3 + a_4$ is $\frac{73}{648}$

38. Let $f(x) = \begin{cases} 3\sin x + a^2 - 10a + 30 & x \notin Q \\ 4\cos x & x \in Q \end{cases}$, then which one of the following statement(s) is/are correct?

A) $f(x)$ is continuous for all x when a = 5.

B) $f(x)$ must be discontinuous for all x, when a = 5.

C) $f(x)$ is continuous for all $x = 2n\pi - \tan^{-1}\left(\frac{3}{4}\right)$, $n \in I$, when a = 5.

D) $f(x)$ is continuous for all $x = 2n\pi - \tan^{-1}\left(\frac{4}{3}\right)$, $n \in I$, when a = 5.

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39. If $f(x)=x^5+x^4+x^3+x^2+x+3$. Then which of the following option(s) is/are correct for the equation $f(x)=0$.

A) Odd number of real roots.

B) Atleast one root lies between $(-\infty,-1)$.

C) Greatest integer value of at least one real root is 2.

D) All real root(s) is/are negative.

40. If equations $x^6+ax^4-4x=0$ and $x^7+ax^5-1=0$, $a\in R$ have a common real root, Then Which of the following option(s) is/are correct?

A) Sum of all possible common real roots is 1.

B) Sum of all possible common real roots is 0.

C) Sum of all possible values of a is $\frac{-1}{2}$.

D) Sum of all possible values of a is 0.

41. Let tangent drawn to the parabola $C_1: y^2=4ax, a>0$ at P meets the y-axis at Q. Another parabola C_2 with vertex Q and focus (0, 0) is drawn which passes through (2a, 0). Then Which of the following option(s) is/are correct?

A) The equation of the curve C_2 can be $x^2=4a(y+a)$.

B) The equation of the curve C_2 can be $x^2=-4a(y+a)$.

C) Point P is the extremity of latus rectum of curve C_1.

D) Possible equation of directrix of curve C_2 can be $y-2a=0$.

42. A and B two $n\times n$ matices such that $\det(A)\neq 0, A+B=(AB)^2$ and $BAB=A+I$. Then Which of the following option(s) is/are correct?

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A) $A^{-1}=\left(A^4-I\right)$ **B)** $B^5-A^5=A+B$

C) $A^9=A^4+A+I$ **D)** $A^2B^2=BA^2B$

43. If $\left|\vec{a}+\vec{b}+\vec{c}\right|=\left|\vec{a}-\vec{b}+\vec{c}\right|=\left|\vec{a}+\vec{b}-\vec{c}\right|=\left|\vec{b}+\vec{c}-\vec{a}\right|$-and if $\left|\vec{a}\right|+\left|\vec{b}\right|=\left|\vec{c}\right|\neq 0$, $\left|\vec{a}\right|=1$ then which of the following option(s) is/are correct?

A) Maximum value of $\dfrac{\left|4\vec{a}+4\vec{b}+3\vec{c}\right|}{\left|\vec{c}\right|}$ is 5.

B) Minimum value of $\dfrac{\left|4\vec{a}+4\vec{b}+3\vec{c}\right|}{\left|\vec{c}\right|}$ is $\sqrt{15}$.

C) Minimum value of $\dfrac{\left|4\vec{a}+4\vec{b}+3\vec{c}\right|}{\left|\vec{c}\right|}$ is $\sqrt{17}$.

D) Maximum value of $\dfrac{\left|4\vec{a}+4\vec{b}+3\vec{c}\right|}{\left|\vec{c}\right|}$ is $3\sqrt{5}$.

44. Let $f(x)=\begin{cases} x[x] & -4\le x<-1 \\ 1-x & -1\le x<1 \\ x^2-3x+2 & 1\le x<2 \\ (x-2)[x] & 2\le x\le 4 \end{cases}$, then (where [.] denotes greatest integer function), Then Which of the following option(s) is/are correct?

A) Number of points where $f(x)$ is non-differentiable is 6.

B) Sum of global maximum value and global minimum value of $f(x)$ is $\dfrac{63}{4}$.

C) Number of points of local extrema of $f(x)$ is 3 points.

D) Number of solutions of $f(x)=4$ is 3.

SECTION – II
(NUMERICAL VALUE TYPE)

This section contains 6 questions. Each question is numerical value type. For each question, enter the correct numerical value **(e.g. 003.2343, 6.250023, 7.000000, -0.33, -.30, 30.27000, -127.0030).**

Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases. Section 2 (Max Marks: 18)

- Section two contains six (6) questions. The answer to each question is a NUMERICAL VALUE
- For each question enter the correct numerical value of the answer using the mouse and the on-screen virtual numerical keypad in the place designated to enter the answer. If the numerical values has more than two decimal places, truncate/round off the value to TWO decimal places.

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Answer to each question will be evaluated according to the following marking scheme
Full Marks: +3 (If only the correct numerical value is entered)
Zero Marks: 0 (In all other cases)

45. If $\int_{1/5}^{5} \frac{1}{x}\left(\left\{\frac{x}{4x^2-2x+9}\right\}+\left\{-\frac{x}{9x^2-2x+4}\right\}\right)dx=\ln(p)$ where $\{.\}$ denotes fractional part function, then the value of p is

46. If $\tan^{-1}\alpha=\tan^{-1}(\alpha+k)+\frac{\pi}{4}$ is true for atleast one $\alpha\in[0,2]$ and $\alpha\in I$, then number of all possible integral value(s) of k is

47. Number of solutions of the set of equations $\sin\theta=\cos 5\theta$ and $\tan 2\theta=\cot 4\theta$ *where* $\theta\in[0,2023\pi]-\left\{\frac{(2n+1)\pi}{8}\right\}-\left\{\frac{(2p+1)\pi}{4}\right\}; n, p\in I$ is

48. If $f(x)=\begin{vmatrix}\sec x & \cos x & \sec^2 x+\cot x\,\mathrm{cosec}\,x\\ \cos^2 x & \cos^2 x & \mathrm{cosec}^2 x\\ 1 & \cos^2 x & \cos^2 x\end{vmatrix}$ then $\int_0^{\pi/2} f(x)\,dx=-\frac{p}{15}-\frac{\pi}{q}$ where p, q are integers, the value of p + q is___

49. A line L on the plane $2x+y-3z+5=0$ is at a distance 3 unit from the point $P(1,2,3)$. A spider start moving from point A and after moving 4 unit along the line $\frac{x-1}{2}=\frac{y-2}{1}=\frac{z-3}{-3}$ it reaches to point P and from P it jump to line L along the shortest distance and then moves 12 unit along the line L to reach at point B. If the distance between the point A and B is λ then $\left[\frac{\lambda}{2}\right]$ equals
(where [.] denotes greatest integer function)

50. Number of complex numbers "z" satisfying $|z(3-4i)+\bar{z}(3+4i)+4|+|z(4-3i)+\bar{z}(4+3i)+6|=10$ and $|z+1|+|z-i|=\sqrt{2}$ is

SECTION-III
(MATCHING LIST TYPE)

This section contains 4 questions, each having two matching lists (List-1 & List-II). The options for the correct match are provided as (A), (B),(C) and (D) out of which ONLY ONE is correct.
Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.
Section 3 (Max Marks: 12)

- This Section contains two (02) List-Match sets.
- Each List-Match set has TWO (02) Multiple-choice Questions
- Each List-Match set has two lists: List 1 and List 2
- List 1 has four entries (I), (II), (III), (IV) and List 2 has six entries (P), (Q), I, (S), (T) and (U)
- Four options are given in each multiple choice question based on List 1 and List 2 and Only ONE of these four options satisfies the condition asked in the Multiple Choice Question.

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Answer to each question will be evaluated according to the following marking scheme
Full Marks: +3 (If only the option corresponding to the correct combination is chosen))
Zero Marks: 0 (If none of the options is chosen (i.e. the question is unanswered)
Negative Marks: -1 (In all other cases)

Answer Q.51 and Q.52 by appropriately matching the list based on the information given in the paragraph

List-I		List-II	
I)	The value of $\left[\frac{2009!+2006!}{2008!+2007!}\right]-2008$ ([.] denotes greatest integer function)	P)	0
II)	If the distance of a point P on the ellipse $\frac{x^2}{6}+\frac{y^2}{2}=1$ from the centre is 2, P has eccentric angle $\theta\in(0,2\pi)$ then the value of $[\theta]$ can be ([.] denotes greatest integer function)	Q)	1
III)	If eccentricity of conjugate hyperbola of the given hyperbola is e' then value of 4e' is, where eccentricity of hyperbola $e=5/3$	R)	2
IV)	Distance of point $P(\alpha,\beta)$ from (2,3), if the tangents to the curve $y^2=4x$ from point P are perpendicular, is	S)	3
		T)	4
		U)	5

51. Which of the following is the only CORRECT combination?

A) $(I),(P)$ B) $(II),(Q)$ C) $(III),(R)$ D) $(IV),(U)$

52. Which of the following is the only CORRECT combination?

A) $(I),(U)$ B) $(II),(T)$ C) $(III),(U)$ D) $(IV),(U)$

Answer Q.53 and Q.54 by appropriately matching the list based on the information given in the paragraph

Consider the following frequency distribution:

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x_i	2λ	3λ	5λ	6λ	7λ
f_i	4	2	3	2	1

Where, λ is a positive integer and the variance of above frequency distribution is 114.
Match each entry in List-I to the correct entries in List-II.

List-I		List-II	
P)	The mean of above data is	1)	10
Q)	The median of above data is	2)	18
R)	The mean deviation about the mean of the above data is	3)	24
S)	The mean deviation about the median of the above data is	4)	8
		5)	4

53. Which of the following is the only CORRECT combination?
A) $P \to 3; Q \to 3$ B) $P \to 4; Q \to 3$ C) $P \to 3; Q \to 4$ D) $P \to 3; Q \to 2$

54. Which of the following is the only CORRECT combination?
A) $R \to 1; S \to 2$ B) $R \to 1; S \to 3$ C) $R \to 2; S \to 2$ D) $R \to 1; S \to 1$

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30 RANK OM PRAKASH BEHERA HT. No. 252021018*	32 RANK SUNKARA SAI RISHANTH REDDY HT. No. 256165327	34 RANK DHRUBA JYOTHI PANJA HT. No. 252048248*	35 RANK BHAVESH JAYANTHI HT. No. 251043080	36 RANK ADVAY MAYANK HT. No. 252104113*	37 RANK KARMANYA GUPTA HT. No. 252081477*
42 RANK MD ANAS HT. No. 252046210*	45 RANK RAMIT GOYAL HT. No. 257001113*	52 RANK MAULIK JAIN HT. No. 252079407*	54 RANK GARV HT. No. 252056188*	59 RANK LARISSA HT. No. 252079071*	60 RANK ARYAN BALABADRULA HT. No. 256132077
63 RANK SAMYAJYOTI BISWAS HT. No. 255058456*	64 RANK AARUSH ANAND HT. No. 251006176*	72 RANK RUSHMITH BANDARI HT. No. 256168048	78 RANK KORIKANA RASAGNYA HT. No. 256057046	87 RANK LAKSHYA SHARMA HT. No. 252070079*	91 RANK AVANEESH BANSAL HT. No. 251113130*	95 RANK KAVYA AGGARWAL HT. No. 252079121*

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